Finding a Way Back: Human–Technology Coupling After the Pandemic

Digital inclusion, reward, avoidance and the gradual reconstruction of participation

K. A. Croft, MA

Abstract

Digital inclusion is commonly understood in terms of access, skills and confidence. These remain necessary, but they do not fully describe the position of someone who is technically capable, continuously connected and nevertheless struggling to participate in work, community or everyday life.

This paper considers a post-pandemic paradox: the same technologies that provide access to support, information and human connection can also become embedded in patterns of avoidance, compulsive checking and prolonged isolation. I explore this through my own experience of neurodivergence, recovery, grief, complex trauma and an ongoing attempt to return to work.

Drawing on reward learning, behavioural activation, exposure theory and metacognition, I propose **titrated reconnection**: a process in which a person chooses a meaningful activity and adjusts its social, sensory and practical demands with appropriate human support. The proposal extends my earlier work on Human–AI Coupling Regulation, or HACR, from the risks of anthropomorphising AI towards the wider relationship between technology use and participation.

My experience is a starting point for inquiry, not evidence that the proposed approach works for others. The paper distinguishes established psychological concepts from a practical model that requires evaluation.

1. The problem of thinking my way out

After lockdown, I believed that if I could understand my situation sufficiently, I could think my way out of it.

My training and interests encouraged that response. I have a background in electronics, design, technology and human–computer interaction, completed two years of undergraduate study in Applied Psychology and Computing, and hold an MA in Sound Design for the Moving Image. I am analytical and comfortable working with complex systems.

During the period in question, I was developing technology and writing a novel about trauma, addiction and systems. The work absorbed me. Isolation also suited aspects of my neurodivergence: it offered control over sensory input, uninterrupted concentration and relief from the effort of masking.

Yet I became increasingly detached from ordinary social participation. I do not mean psychosis or a loss of reality testing. I mean that I could spend long periods inside, writing and developing, while contact with people and returning to work became progressively harder.

My father died during this time. I was also working through complex trauma, including CPTSD, experiences associated with addiction before recovery, and other earlier experiences. There is no defensible single explanation for how those circumstances interacted.

I had friends and support. I also had more than twenty-five years of experience in twelve-step recovery. But recognising an experience intellectually did not automatically give me the capacity to change the conditions in which it was occurring.

This distinction between **insight and action** is central to the argument. Understanding a feedback loop can be necessary without being sufficient to interrupt it.

2. Post-pandemic participation and digital exclusion

During lockdown, activities that had depended on physical presence moved online. For me, Zoom and online recovery groups provided meaningful contact and support. They still do.

The difficulty emerged when physical spaces reopened but my own participation did not simply resume. A situation could be objectively available and subjectively difficult to enter.

Consider a library. I know I am entitled to go in, sit down and read. Yet I can still experience the building as an institution in which I might need permission to remain. Information about opening hours addresses access to the building; it does not necessarily address the expectation of being unwelcome.

The wider employment context makes these questions consequential. The ONS estimated that, in June–August 2026, **474,000 people aged sixteen and over had been unemployed for more than twelve months**. In the same period, approximately **2.74 million people aged sixteen to sixty-four were economically inactive because of long-term sickness**. These are different categories: unemployment generally concerns people actively seeking and available for work, while economic inactivity includes people outside those conditions. ONS, October 2026

In April–June 2025, the UK employment rate for disabled people aged sixteen to sixty-four was **52.8%**, compared with **82.5%** for non-disabled people. DWP

Locally, the Dorset and Bournemouth, Christchurch and Poole working plan identifies **28,300 people economically inactive because of long-term sickness across the two council areas in 2024/25**. Dorset Council

England's 2023/24 Adult Psychiatric Morbidity Survey found common mental health conditions among **22.6% of people aged sixteen to sixty-four**, compared with 18.9% in 2014. The change cannot be attributed to the pandemic alone. NHS England

These populations overlap but are not interchangeable. Long-term unemployment includes people without a diagnosis; disabled people are not necessarily unable to work; and long-term sickness includes physical as well as mental health conditions.

The UK's digital inclusion policy emphasises access, skills, support and confidence. Digital Inclusion Action Plan I suggest that it is also necessary to consider **how technology use relates to a person's capacity to act on their own goals**.

A person may be digitally proficient and still experience social or economic exclusion. Equally, a person can lack access to appropriate devices or skills. Both situations matter.

3. From HACR to the wider digital loop

My initial HACR work was concerned with human–AI coupling: how interaction between a person and an AI system can produce risks that neither side explains alone.

Over approximately four years of using AI, I recognised that I was anthropomorphising it. I could become angry when it did not do what I wanted, as if it understood my intention and was refusing to cooperate.

Using meta-prompting—questioning the prompts, assumptions and interaction itself—I began examining the system's limitations and my own responses. AI could produce hallucinations, confident errors, contradictions and context loss. It could also appear to understand while failing to retain what mattered.

That process required caution. An AI system's account of its own behaviour may itself be inaccurate. NIST identifies confabulation, overreliance and risks arising from human–AI interaction as concerns in generative AI systems. NIST AI Risk Management Framework: Generative AI Profile

The further question is whether a useful interaction can also become part of an avoidance pattern.

A difficult employment decision might lead me into researching a technical solution. That produces insight, structure and an immediate sense of progress. But if the work repeatedly postpones contact with an employer, collaborator or community, productivity and avoidance may coexist.

This is not a judgement that technical work is unreal or worthless. The question is whether the interaction remains connected to the purpose I intended it to serve.

4. Reward, dopamine and potentially addictive use

Digital platforms can offer frequent cues and uncertain outcomes: a notification, a reply, a new post, a useful answer or a development breakthrough. Each can invite another check.

Reinforcement learning provides one way to understand why behaviour may recur. Actions are influenced by their consequences, including rewards and the effort needed to obtain them. In a study of more than one million social-media posts, Lindström and colleagues found that posting patterns were consistent with reward-learning models; an experiment also found that social feedback influenced subsequent behaviour. This does not establish that every user is addicted. Lindström et al., 2021

Dopamine is relevant, but the familiar description of it as simply a "pleasure chemical" is inadequate. Dopaminergic systems are involved in learning, motivation and responding to salient cues. Schultz and colleagues' work on **reward prediction error** describes signals associated with differences between expected and actual rewards. Schultz, Dayan and Montague, 1997

Berridge and Robinson distinguish **wanting**—the motivational pull of a cue—from **liking**, the pleasure obtained from an experience. A person may repeatedly want to check a feed without finding much satisfaction in doing so. Their account was developed substantially through addiction research; it should not be treated as proof that ordinary scrolling produces the same neurological changes as substance dependence. Berridge and Robinson

This gives the digital problem greater precision. A cue may acquire motivational significance; checking becomes familiar; the response can offer novelty, reassurance, distraction or relief. Repetition may continue even when the activity no longer serves the person's broader intentions.

That is different from claiming that scrolling "burns out dopamine pathways". The latter is not a conclusion supported by the evidence presented here.

Nor should all extensive use be called addiction. The World Health Organization recognises **gaming disorder** where impaired control, priority given to gaming and continuation despite harm produce significant functional impairment, normally over at least twelve months. This does not make every form of heavy screen use a diagnosable disorder. WHO

A 2026 meta-analysis of adolescents and young adults found that problematic social-media use was associated with loneliness more strongly than time spent using social media. The studies varied substantially, and association does not establish whether use causes loneliness, loneliness contributes to problematic use, or both reflect other factors. Feng et al., 2026

5. Avoidance, relief and the cost of alternatives

A digital loop need not be sustained by pleasure alone. It can also be maintained by **negative reinforcement**: an action is repeated because it reduces an uncomfortable feeling in the short term.

If approaching a library, making a telephone call or contacting an employer produces anxiety, remaining at home may bring immediate relief. The relief makes avoidance understandable. But repeated avoidance can reduce opportunities to discover what would happen under different conditions.

The competing activity also has a cost. A social outing may involve travel, uncertainty, sensory overload, fatigue or fear of judgement. An online activity may be immediately available, familiar and controllable. From that position, choosing the screen is not mysterious.

This is why asking someone simply to “use technology less” is inadequate. It may remove a coping mechanism or a source of support without making another form of participation more accessible.

In a small observational study of 72 young adults, occasions when alternative activities were perceived as more rewarding were followed by lower reported problematic smartphone use. The finding is suggestive, but the sample was narrow and the study does not prove that providing alternatives would cause the same effect for other groups. Ecological momentary assessment study

A separate randomised trial found improvements in wellbeing, mental health and sustained attention when participants were assigned to block smartphone internet access for two weeks while retaining calls, texts and internet access on other devices. Adherence was difficult: only **119 of 467 participants** met the researchers’ compliance criterion. It was not a test of an addiction treatment or of Happyish’s proposed model. Castelo et al., 2025

The implication I draw is not that everyone should disconnect. It is that **selective boundaries and accessible alternatives deserve investigation together**.

6. Sensory access, desensitisation and titration

Access to Work funding gave me equipment, including a good MacBook. High-quality noise-cancelling earbuds became a particularly significant adjustment: they changed my life.

Loud noises, bangs and competing background sound are difficult for me. They seemed more difficult after the pandemic. I suspect that, before isolation, regular contact with those environments had made some of them more familiar or manageable. That is my interpretation, not proof of exactly what changed neurologically.

A useful psychological distinction is between a **feared expectation** and a **sensory barrier**. “I may be judged if I enter the library” is a different problem from “the sound in the library is overwhelming.” They may interact, but they do not require identical responses.

Noise-cancelling earbuds can make a setting accessible enough for participation. In research on autistic traits and auditory environments, many participants reported using sound-reducing technologies; the study supports taking sensory experience and accommodation seriously, although it does not prove the effect of earbuds for every individual. Auditory environments study

Systematic desensitisation combines gradual exposure with relaxation. **Graded exposure** approaches feared situations in stages. These are clinical concepts with specific applications. American Psychological Association

Contemporary exposure research also examines **inhibitory learning**: a person may acquire new information that competes with an earlier feared expectation, rather than simply erasing the old association. If I expect to be challenged for sitting in a library and instead sit undisturbed, the experience may provide information my anticipation did not. One uneventful visit does not guarantee that fear will disappear or generalise to every setting. Craske et al., 2014

I use **titration** to describe the ongoing adjustment of practical demands: duration, complexity, sensory load, social intensity and support. It may mean visiting briefly with a friend and earbuds, repeating the same visit, or deciding that another activity is preferable.

An accommodation should not automatically be treated as a barrier to progress. My proposal concerns supported participation; it does not ask people to abandon necessary adjustments.

7. Action, reflection and the role of other people

What improved my confidence was often mundane: meeting somebody to do something, sitting in a library with a companion, or going to a shop twice a week with friends—even when I did not know what the hell to talk about.

I also use live YouTube ADHD companion or “body-doubling” sessions while working. Another person’s presence, even mediated through a screen, can change the experience of beginning and continuing a task.

These activities connect with **behavioural activation**, which examines relationships between behaviour and mood and can involve reintroducing meaningful activities. Clinical behavioural activation is a structured treatment; my ordinary activities are not evidence that I received or delivered that treatment. NICE guidance

Repetition gave me more opportunities to discover that I could participate. My confidence has improved, and my social anxiety is considerably better. Companionship, sensory adjustments, familiarity and other aspects of my circumstances may all have contributed.

Recovery also gave me a capacity to step back and examine my patterns. **Metacognition** describes awareness and evaluation of one’s own thinking. In my fellowships, examining behaviour is also relational: friends can notice a pattern I have missed, or reflect an ability I have discounted.

I have been using myself as a test guinea pig in my attempt to return to work. That makes my experience a source of questions, not a controlled experiment.

There is a further hypothesis I want to explore separately: prolonged solitude may reduce opportunities for some people to have both helpful and unhelpful patterns reflected back to them. That is not a claim that solitude prevents growth. It is a question about feedback, context and whether a person has access to perspectives outside their own.

8. Shame, parts and self-parenting

Honesty matters to me because it makes the relationship between shame and participation visible.

A person who has been out of work for years may be asked to explain a gap, a health condition or a change in confidence. The practical task can become entangled with how they feel about themselves.

Brené Brown's presentations Listening to Shame and The Power of Vulnerability consider shame, vulnerability and connection. For me, the relevant distinction is between examining a behaviour honestly and treating that behaviour as proof of personal worthlessness.

For approximately six years I have also undertaken work in a twelve-step fellowship concerned with internal family dynamics and self-parenting. The idea that a person can learn to become a loving parent to themselves has been significant for me.

Internal Family Systems describes a person's experience in terms of parts and a distinct organising Self. **Transactional Analysis** uses a different model of Parent, Adult and Child ego states, including a nurturing Parent. These are frameworks for understanding experience; their categories should not be conflated or presented here as individually established brain structures. IFS Institute, International Transactional Analysis Association

Gabor Maté's work is also relevant. A physician whose background includes family medicine, palliative care and work with people experiencing addiction and mental illness in Vancouver, he considers addiction in relation to trauma and lived context. Maté's background

These ideas help me consider why one part of me may seek safety in seclusion while another wants connection and work. They do not imply that Happyish provides psychotherapy.

9. Titrating technology and habits

I still have soothing digital routines. Before bed I sometimes look at X—I still call it Twitter.

I use an orange or warmer phone setting and dark mode in the evening. During the day I deliberately switch to light mode and a cooler, bluer appearance, which I associate with alertness. I also use mindfulness and other familiar practices to help regulate myself.

These are personal choices, not a prescribed protocol. Light influences circadian rhythms, but a particular phone setting cannot be assumed to produce the same effect for everyone; timing and brightness also matter. National Institute of General Medical Sciences

The broader process has been one of repeatedly adjusting habits and conditions.

James Clear's Atomic Habits is a very good book on the relationship between small actions and the systems around them. Mark Manson's The Subtle Art of Not Giving a F*ck is another very good book for thinking about what deserves attention.

For me, neither insight nor habit change stands alone. A small action becomes more possible when the environment, sensory demands and human support are also considered.

10. A proposal for supported reconnection

Happyish proposes free, person-led digital mentoring that begins with a real task. **Titrated reconnection** extends that approach through six connected steps:

1. **Choose a meaningful goal.** This might concern a service, friendship, community activity, learning or work.
2. **Identify the barriers.** Consider digital access, sensory conditions, health, fatigue, anxiety, shame, practical arrangements and patterns of technology use.
3. **Agree a manageable step.** Adjust its size and conditions with the person, including their right to pause or change direction.
4. **Use technology for a defined purpose.** A screen might support a message, route, appointment or application. AI remains optional, with important information checked.
5. **Support participation.** The intended activity may be online or in a physical setting. Online friendships and recovery groups are human connection.
6. **Review and adjust.** Discuss what happened, whether the task served its purpose, and what the person wants to repeat or change.

For someone concerned about compulsive use, the process might include direct links, fewer notifications or an agreed stopping point, while preserving access to valued communities. Where addiction-related harm requires treatment, specialist support is necessary.

For someone seeking work, the chosen task might connect to a portfolio, training, an application or a conversation with an employer. It must also account for structural conditions: reasonable adjustments, accessible recruitment, transport, suitable hours and sustainable work.

11. What would need to be tested

The proposed model brings together practical digital inclusion, human contact, sensory accommodation and titration. Its central hypothesis is that these elements can help some people translate digital capability into participation that matters to them.

An evaluation should examine completion of **person-chosen tasks**, perceived agency over technology, social participation, accessibility and—where the person wants it—movement towards suitable work. It should record difficulties and unintended consequences as well as successes.

Fewer screen minutes would not by itself establish improvement. Nor would an employment outcome capture the value of someone reconnecting with a community they chose.

My experience offers a starting point: thinking helped me identify patterns; recovery and friends helped me reflect; technology provided both support and opportunities for avoidance; sensory adjustments made participation more possible; and repeated ordinary activities helped rebuild confidence.

The wider proposition remains to be tested: that a person does not have to choose between digital connection and a life beyond the screen. With appropriate boundaries, adjustments and human support, technology can become part of a route towards the participation they choose.